

Origins and Evolution of Language

Week 3: Intention and structure in animal communication

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An in-person question from last week

“Are the exam questions just on the lecture and tutorial readings? Or will there be additional examinable stuff said in the lectures?”

“What kind of questions will be present in the exam, and how much weight will different types of questions have?”

“Generally, what parts of information should be kept in mind or note and particularly pay attention to in weekly readings?”

A question from the pre-lecture quiz (from someone who has noticed there are **no classes next week**)

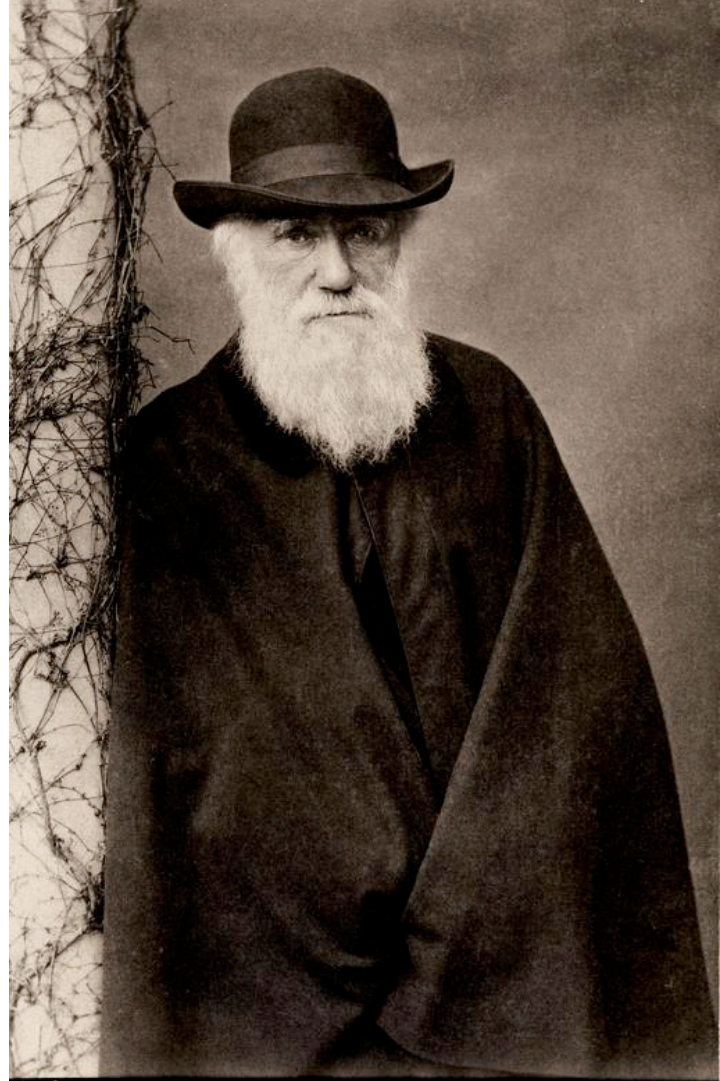
“Although we are still a couple of months away from the exam, on reading week is there anything specific that we should revise? Or should we just continue to go over the previous weeks’ slides/readings/tutorial exercises etc.”

Plan for today

- Finishing up on last week
 - Because I forgot I can go until 4pm 🙄
- Brief summary of Fitch (2019)
- Spotlight on **intentional** communication in primates
- Spotlight on **structure** in primates and birds
- Spotlight on **learned communication** in primates and birds

Language as an adaptation?

- Language is just like any other **adaptive** feature of an organism's biology
- It's an **innate** feature of the human mind
- It **evolved by natural selection** under pressure for communication



A tool for the communication of knowledge and internal states



The core of the argument (from Pinker & Bloom, 1990)



“We will suggest that language shows signs of design for the communication of propositional structures over a serial channel.

The propositions [in human thought] are relational structures whose symbols pertain to people, objects, and events, the categories they belong to, their distribution in space and time, and their causal relations to one another ... The causal relations governing the behavior of other people are understood as involving their beliefs and desires, which can be reconsidered as relations between an individual and the proposition that represents the content of that belief or desire”

Pinker, S., & Bloom, P. (1990). Natural language and natural selection.
Behavioural and Brain Sciences, 13, 707-784.
Lengthy quote is from p. 712-713

The core of the argument (from Pinker & Bloom, 1990)



“We would want to be able to

- refer to individuals and classes,*
- to distinguish among basic ontological categories (things, events, places, times, manners, and so on),*
- to talk about events and states, distinguishing the participants in the event or state according to role (agents, patients, goals),*
- to talk about the intentional states of ourselves and others*
- to express distinctions of truth value, modality (necessity, possibility, probability, factivity)*

Pinker, S., & Bloom, P. (1990). Natural language and natural selection.
Behavioural and Brain Sciences, 13, 707-784.
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The core of the argument (from Pinker & Bloom, 1990)



- *to comment on the time of an event or state, including both its distribution over time (continuous, iterative, punctate) and its overall time of occurrence*
- *to encode an unlimited number of predicates, arguments, and propositions*
- *to be able to use the same propositional content within different speech acts; for instance, as a question, a statement, or a command.*
- *to focus or to put into the background different parts of a proposition, so as to tie the speech act into its context of previously conveyed information and patterns of knowledge of the listener. ”*

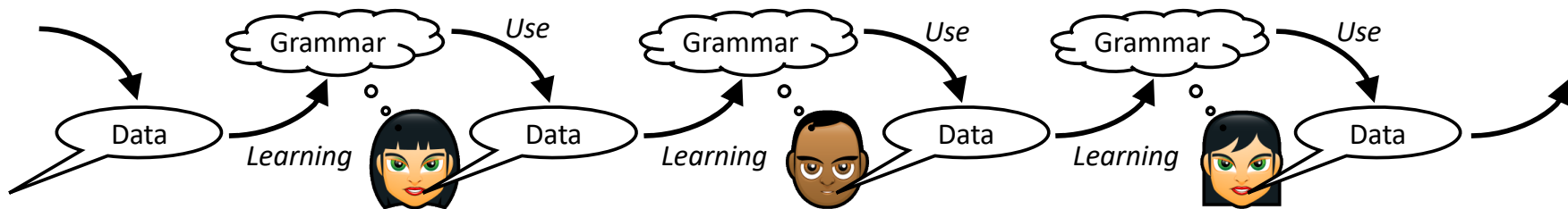
The core of the argument (from Pinker & Bloom, 1990)



“All we have argued is that human language, like other specialized biological systems, evolved by natural selection. Our conclusion is based on two facts that we would think would be entirely uncontroversial: Language shows signs of complex design for the communication of propositional structures, and the only explanation for the origin of organs with complex design is the process of natural selection.” (p. 726)

“Going over question 8 [from week 2 pre lecture quiz] in the lecture could help. I am not sure I understood it correctly”

Question 8: [This is an exam-style question] If language is an adaptation, a product of evolution by natural selection, it must meet the requirements for evolution by natural selection - it must exhibit **heritable variation in fitness**. Drawing on the arguments in Pinker (2003) [or Pinker & Bloom (1990)], or on your own evidence, to what extent do you think language exhibits heritable variation in fitness? 1 paragraph max please!



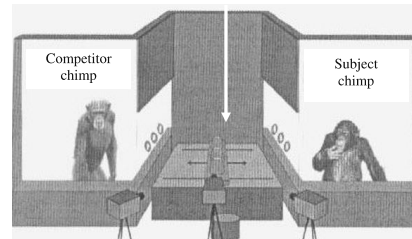
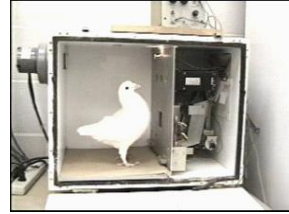
A question from the pre-lecture quiz based on something I probably said last week

“Whether it is really true that human language can meaningfully express any concept we think or feel. I find the claim a little anthropocentric, and I would argue that if it were really the case, misunderstandings would only ever be the fault of the listener, not the speaker. How often do we not trip over our words?”

Summary of Fitch (2019)

Non-humans have rich mental lives...

- Concepts and categories
- Memory and planning
- Hierarchically-structured behaviours
- Tool use
- Knowing what others know
- ...



Alex the parrot



...but their communication systems seem *relatively* restricted

- In the things they communicate about
- ‘Innate’ signal repertoires
 - Particularly among primates
 - But see later today
- “Functionally referential”
 - But not intentional (?)
- Complex vocalisations
 - But less obviously in primates (?)
 - And complexity not subserving meaning



A question from the pre-lecture quiz

“How do you register a concept, and indeed think, without using language?”

A question from the pre-lecture quiz

“What are first and second-order beliefs?”

How exactly did the researchers test self-doubt in non-human animals? And how did they conduct the experiment for same/different colours in honeybees?”

A question from the pre-lecture quiz

“What are first and second-order beliefs?”

First order: having some belief about the world

Second order: meta-cognition, beliefs about your beliefs (e.g. confidence that they are true)

A question from the pre-lecture quiz

“How exactly did the researchers test self-doubt in non-human animals?”

Dautriche, I., Goupil, L., Smith, K., & Rabagliati, H.(2022). Two-year-olds' eye movements reflect confidence in their understanding of words. *Psychological Science*, 33, 1842-1856.



A question from the pre-lecture quiz

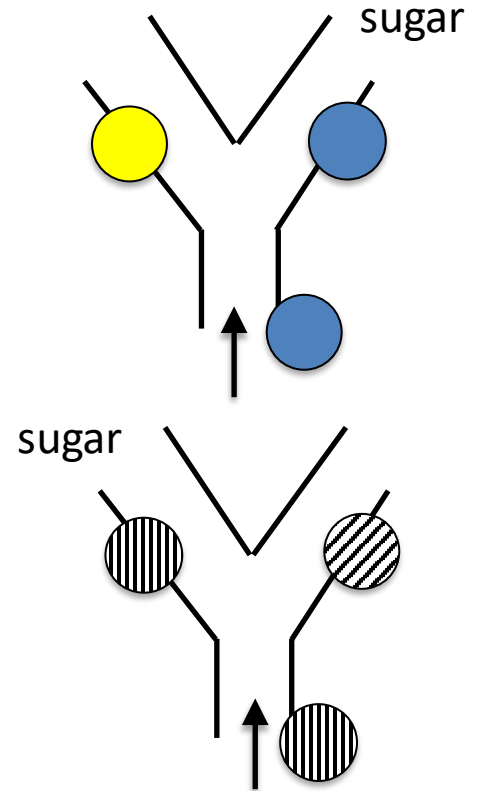
“And how did they conduct the experiment for same/different colours in honeybees?”

“a honeybee can judge whether two stimuli are the same or different in colour or pattern [54]” (Fitch, 2019, p. 5)

A question from the pre-lecture quiz

Giurfa, M., Zhang, S., Jenett, A., Menzel, R., & Srinivasan, M. V. (2001). The concepts of 'sameness' and 'difference' in an insect. *Nature*, 410, 930-933.

"We trained honeybees, *A. mellifera*, in a delayed matching-to-sample paradigm to examine whether they could form a concept of sameness. Training was carried out using a Y-maze placed close to a laboratory window. Each bee entered the maze by flying through a hole in the middle of an entrance wall. At the entrance, the bee encountered the sample stimulus. The sample was one of two different stimuli, A or B [e.g. blue or yellow disk]. The entrance led to a decision chamber, where the bee could choose one of two arms. Each arm carried either stimulus A or stimulus B as secondary stimulus. The bee was rewarded with sucrose solution only if it chose the stimulus that was identical to the sample. If the bees managed to learn the original discrimination, they were presented with new sample and secondary stimuli in 'transfer tests': the bees had to choose between stimuli C and D [e.g. horizontal vs vertical lines], when the sample was either C or D."

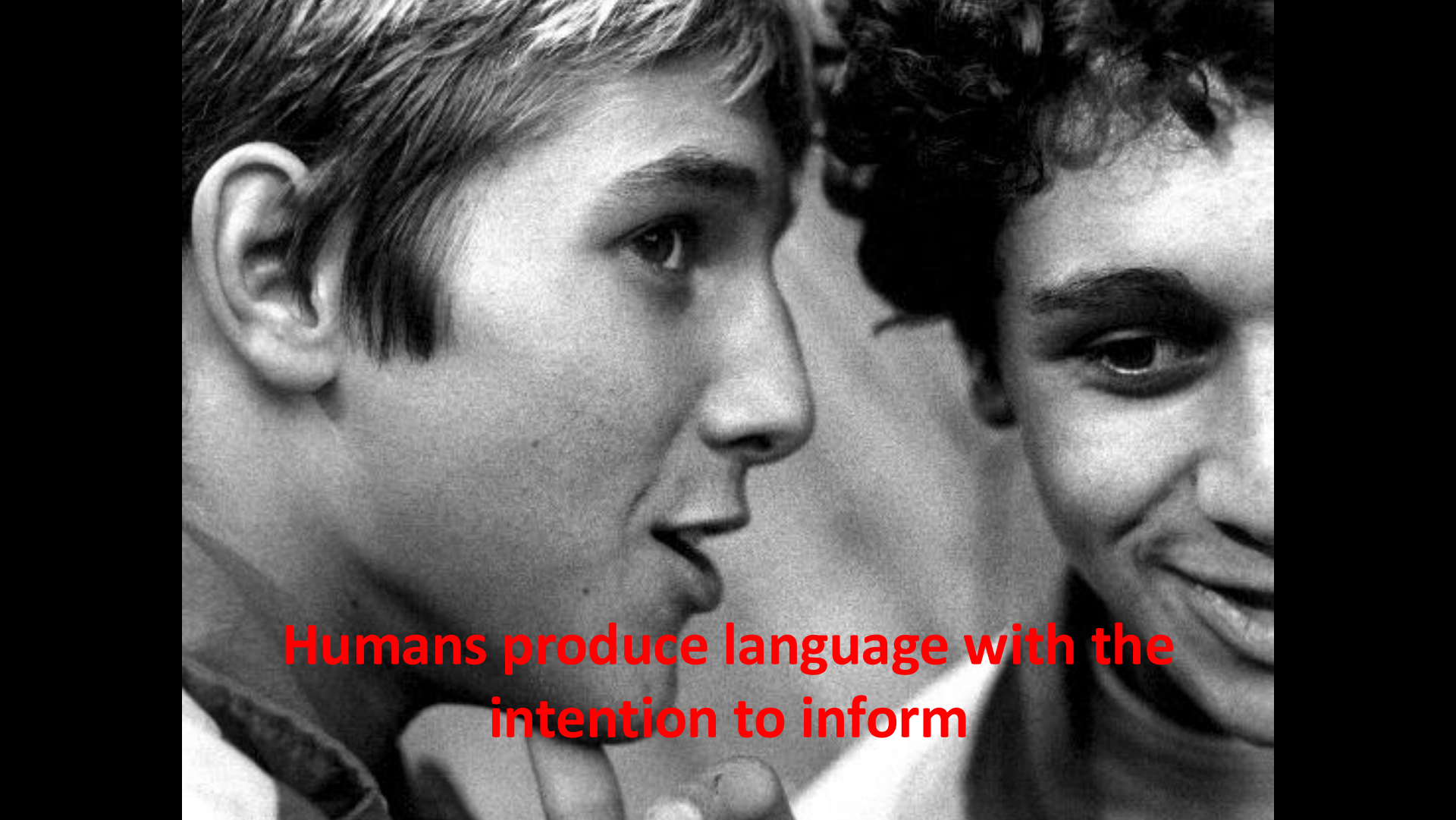


A question from the pre-lecture quiz

“How did humans overcome the limitations of earlier, less flexible communication systems and develop language as we know it today?

Why did non-human primates not undergo a similar development, and could they potentially evolve their own language systems over a long period of time?”

Intentional communication in primates

A black and white photograph showing two young men from the chest up. The man on the left is in profile, facing right, with his mouth slightly open as if speaking. He has short, light-colored hair. The man on the right is also in profile, facing left, looking towards the first man. He has dark, curly hair and is smiling slightly. The background is dark and out of focus.

**Humans produce language with the
intention to inform**



First order intentionality: desire to
modify the **behaviour** of another



Second order intentionality: desire to
modify the **mental state** of another

Functionally referential communication in primates

Can Monkeys Talk?

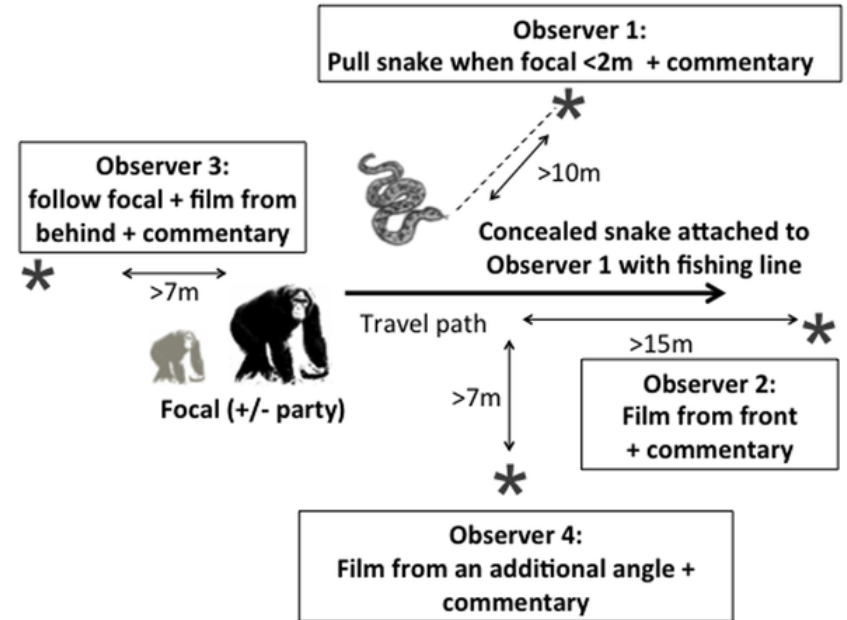
Absence of intentional communication in macaques?

- Mothers and infants
- **Ignorance condition:** Mother knows something, infant doesn't
 - Presence of food, predator
- **Knowledge condition:** They both know it
- **Mothers' vocalizations didn't differ between conditions**



Intentional communication in chimpanzees?

- Wild chimps
- Surprised with snake model, either alone or in part of group
 - Presence of others matters?
 - Gaze-alternation?
 - Persist until others safe?







Structure in primate and avian
communication

Learning in primate and avian
communication

Reminder: structure in language

Inventory of meaningless units
(10s)



Inventory of meaningful units
(1000s)



Inventory of meaningful sentences
(∞)

p t d s ʔ k ɔ ə a ...

ə ʔə -əd dɔg kat ʔat spɔt ...
(a) (the) (past tense) (dog) (cat) (that) (spot)

the cat spotted the dog a dog spotted the cat
a cat spotted the dog the dog spotted the cat
the cat spotted the cat that spotted a dog ...
the dog spotted the cat that spotted the dog

Song in gibbons



Raemaekers, J. J., Raemaekers, P. M., & Haimoff, E. H. (1984). Loud calls of the gibbon (*Hylobates lar*): repertoire, organization and context. *Behaviour*, 91, 146–189.

Putty-nosed monkey

Pyow = leopard

Hack = eagle

Pyow-hack = move

wooclap.com, code INMRBS

What does this system share with language?

1. Not enough to make a comparison
2. Combinatorial phonology
3. Compositional syntax



Campbell's monkey

Leopard alarm



Eagle alarm

Boom = not urgent

wooclap.com, code INMRBS

What does this system share with language?

1. Not enough to make a comparison
2. Combinatorial phonology
3. Compositional syntax

Chimpanzees

Alarm huus: alarm or threat

Waa bark: recruitment call (?)

Alarm huu + waa bark = recruitment signal in
a dangerous situation (?)

Very similar combination reported in
southern pied babblers

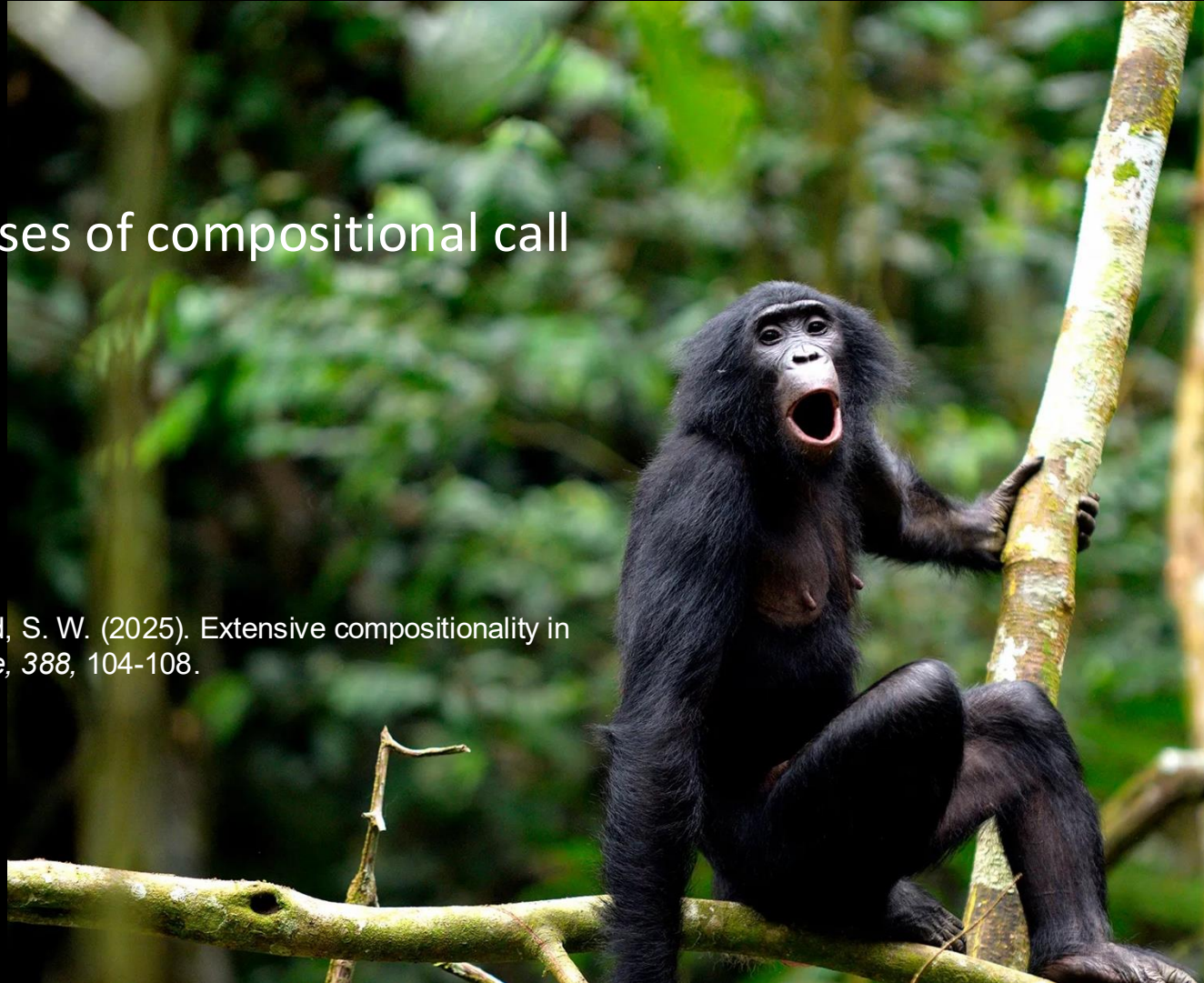
Leroux, M., Schel, A.M., Wilke, C. et al. (2023) Call combinations and compositional processing in wild chimpanzees. *Nature Communications*, 14, 2225.

Engesser, S., Ridley, A. R., & Townsend, S. (2016). Meaningful call combinations and compositional processing in the southern pied babbler. *Proceedings of the National Academy of Sciences, USA*, 113, 5976-5981.

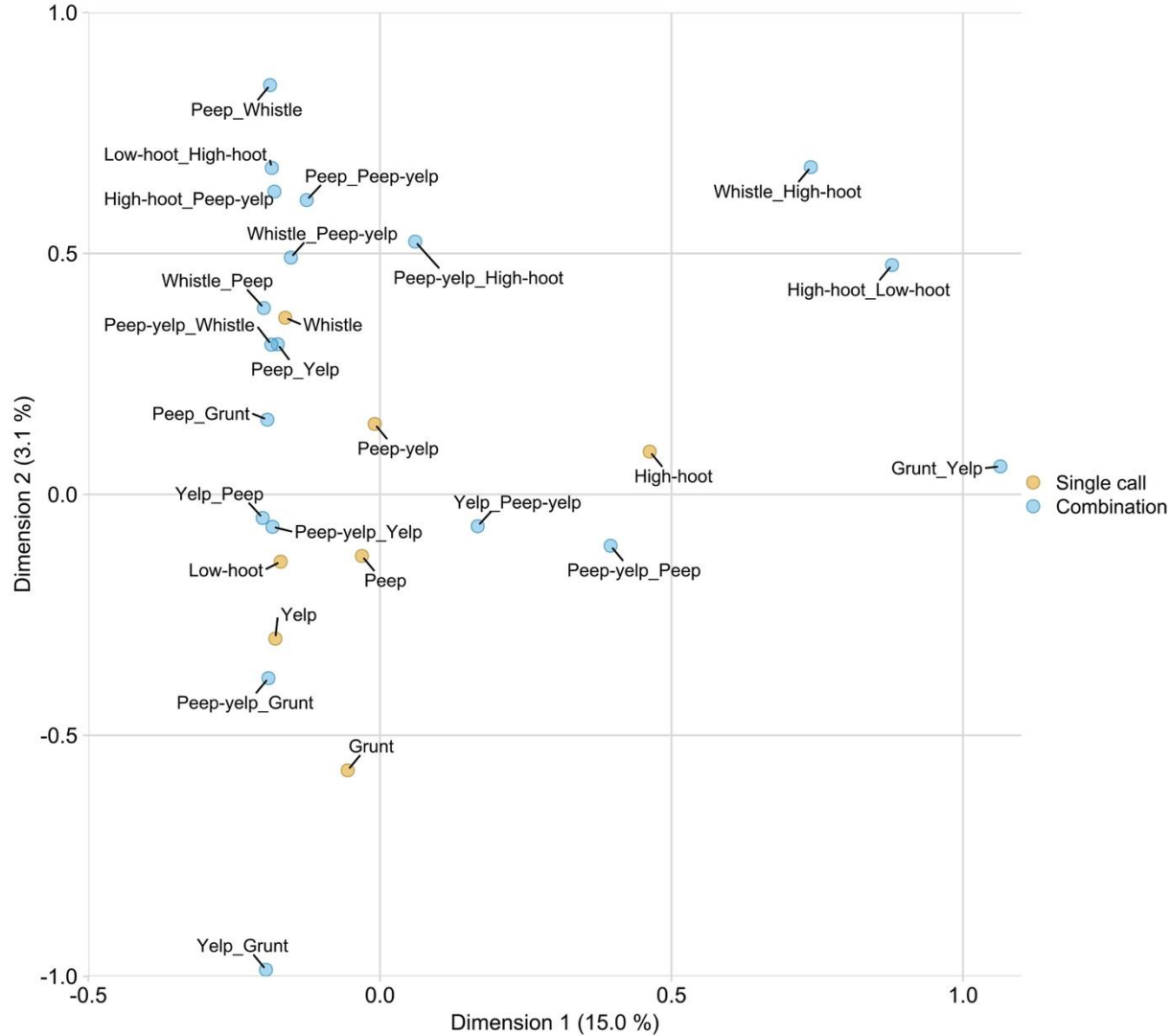
Bonobos?

Multiple similar cases of compositional call combination?

Berthet, M., Surbeck, M., & Townsend, S. W. (2025). Extensive compositionality in the vocal system of bonobos. *Science*, 388, 104-108.



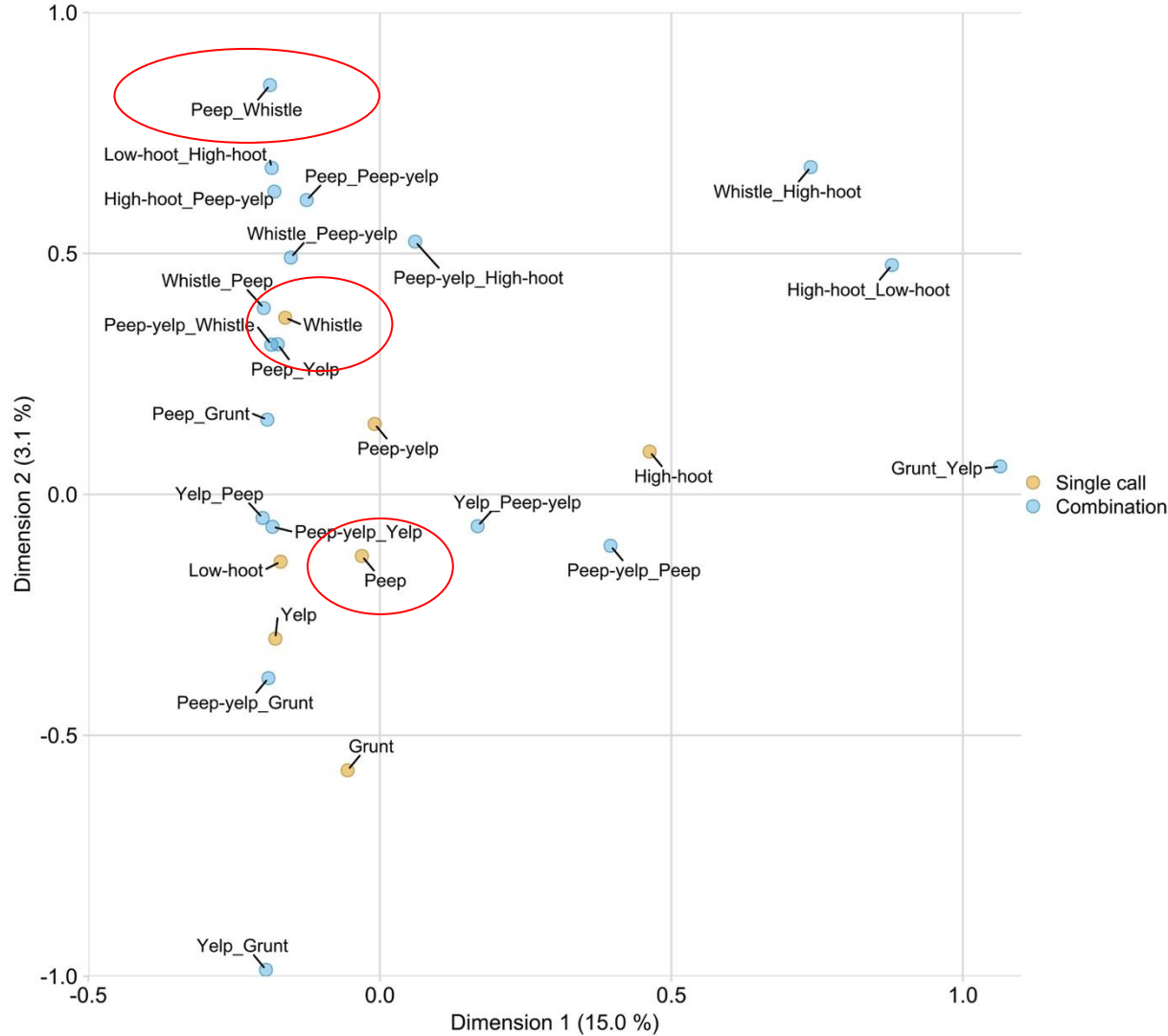
Video illustrating the reaction to the
playback of an “*alarm-hoo+waa-bark*”
combination.





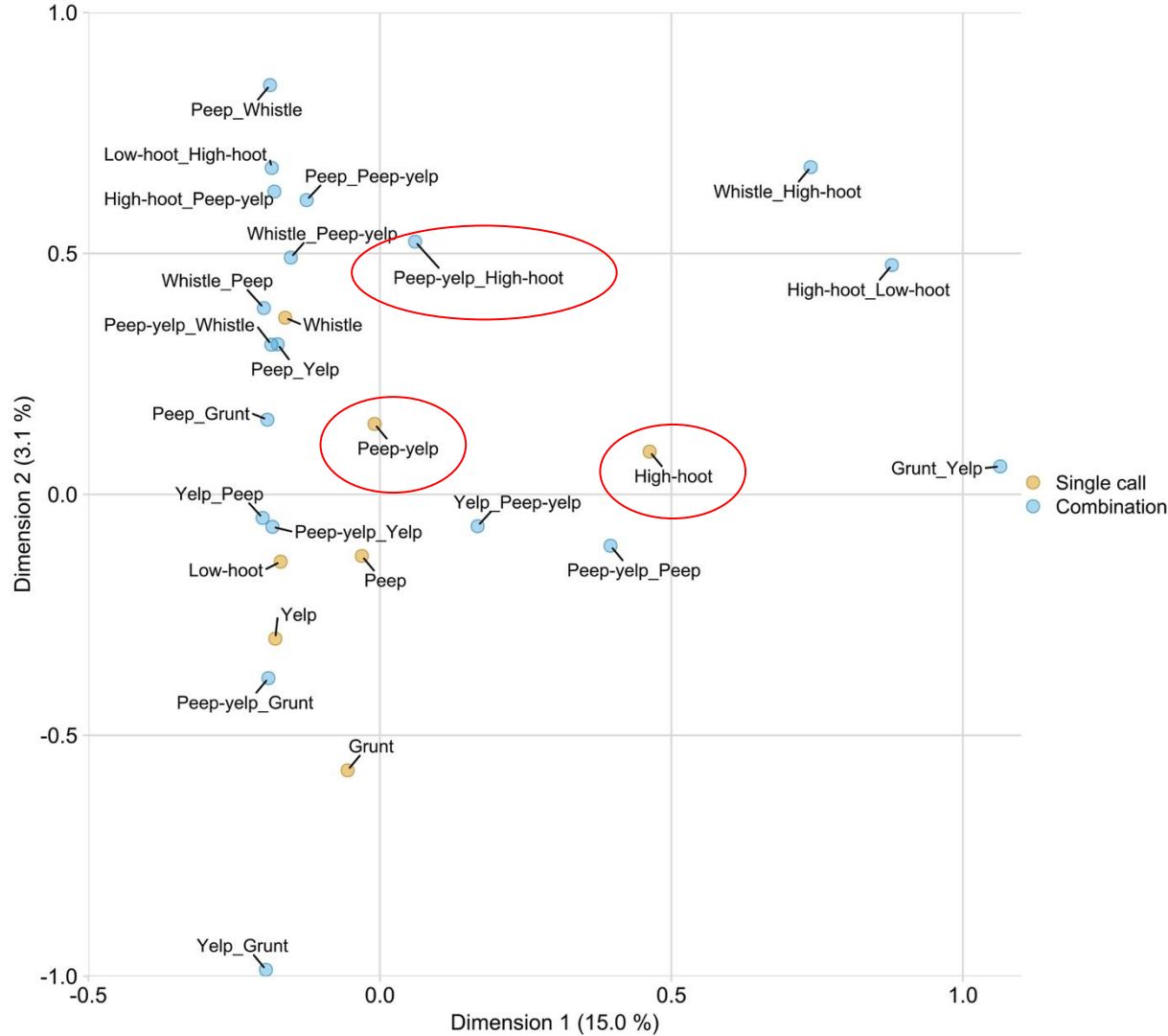
Candidate AB

- Do A and B have different meanings?
- Does AB differ from meaning of A, B?
- Is AB related to the meanings of A, B?
- Is AB different from A+B?



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Bonobos?

Multiple similar cases of compositional call combination?

Their correlational method seems highly prone to false positives (false positive rate as high as 80%!)

Berthet, M., Surbeck, M., & Townsend, S. W. (2025). Extensive compositionality in the vocal system of bonobos. *Science*, 388, 104-108.

Wartel, A. et al. (2026). Structure without semantics: no evidence for bonobo compositionality in Berthet et al. (2025). *PeerJ*, 14, e21651.

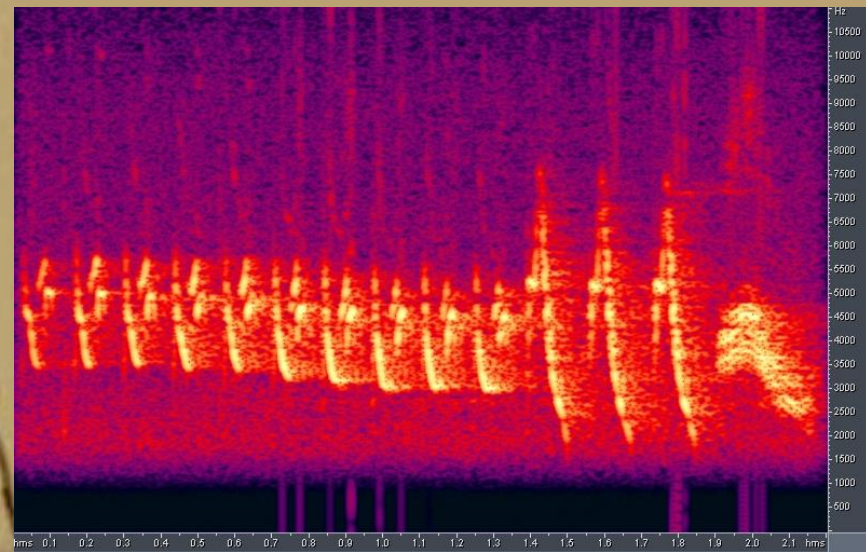


Abundant evidence of structure in **bird song**

- Songs consist of sequences of notes
- Constraints on the order of combination
- Structure in the signal doesn't subserve meaning
- **Vocal learning**
 - Absent in primate vocal behaviour (?)
- Ultimate functions
 - Territorial defense
 - Courtship
 - Pair/group bonding (duetting)



Chaffinch song



Structure of chaffinch song (British)

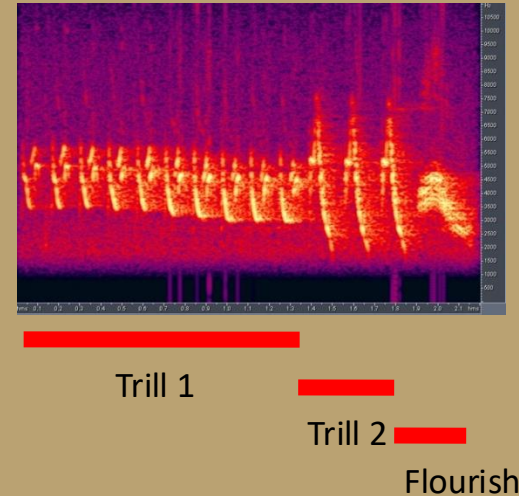
Each bird has 1-6 song types

- Mean 2-3

Order of notes in each song fixed

2-5 trill phrases, followed by a flourish

- Trill: sequence of 2 or more near-identical units
 - Number of repetitions can vary
- Flourish: no repetition
- Transitional notes: single notes between trill phrases
- Re-use of notes
 - Different songs may share, e.g., a flourish



Willow warbler song

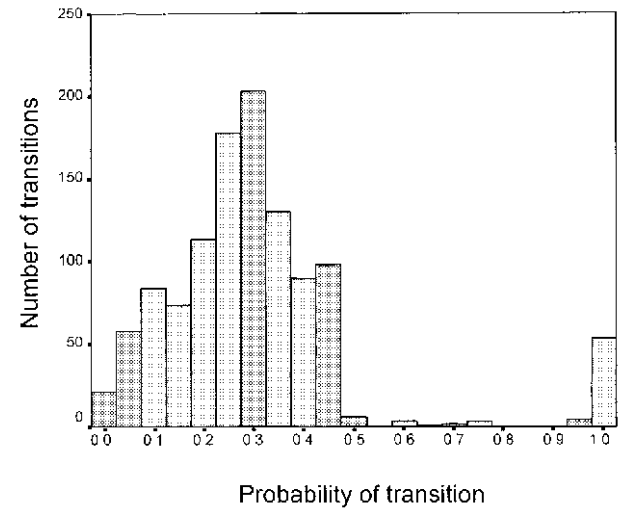


> 100 songs for some birds

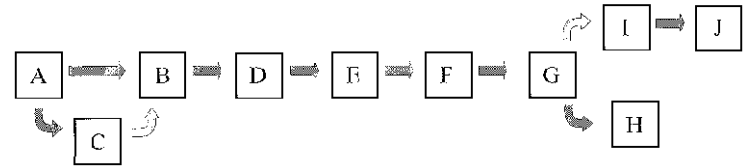
- Repertoire size varies

Mix of predictable and less predictable transitions

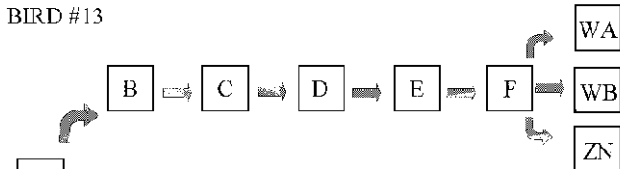
- A simple grammar



BIRD #117



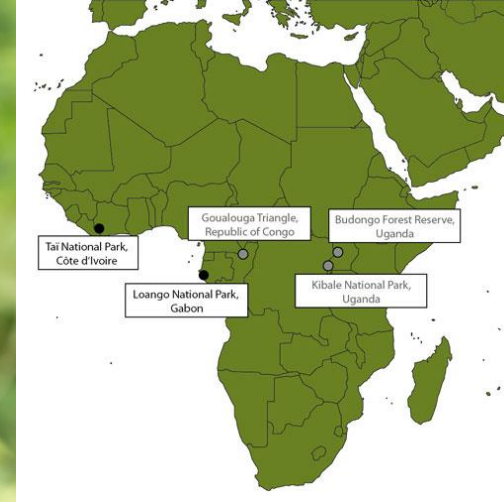
BIRD #13





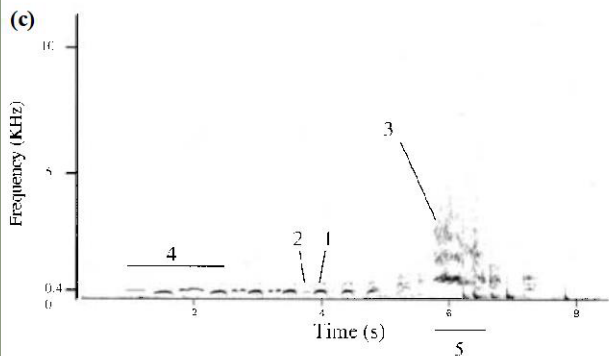
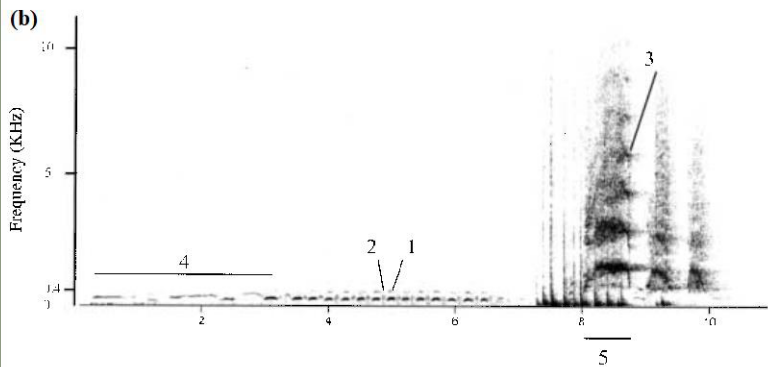
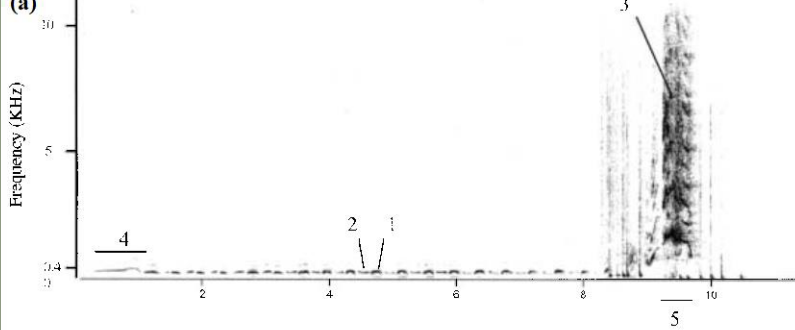
Janie Goodall

Suggestive evidence for learned vocalizations in chimpanzees?



Pant hoots of chimpanzees vary between neighbouring groups

Crockford, C., Herbinger, I., Vigilant, L. & Boesch, C. (2004). Wild Chimpanzees Produce Group-Specific Calls: a Case for Vocal Learning? *Ethology*, 110, 221–243.



Crockford et al. (2004): pant hoots of neighbouring groups differ in (e.g.):

- Length of intro (4)
- Peak frequency of screams (3)
- Duration of climax (5)

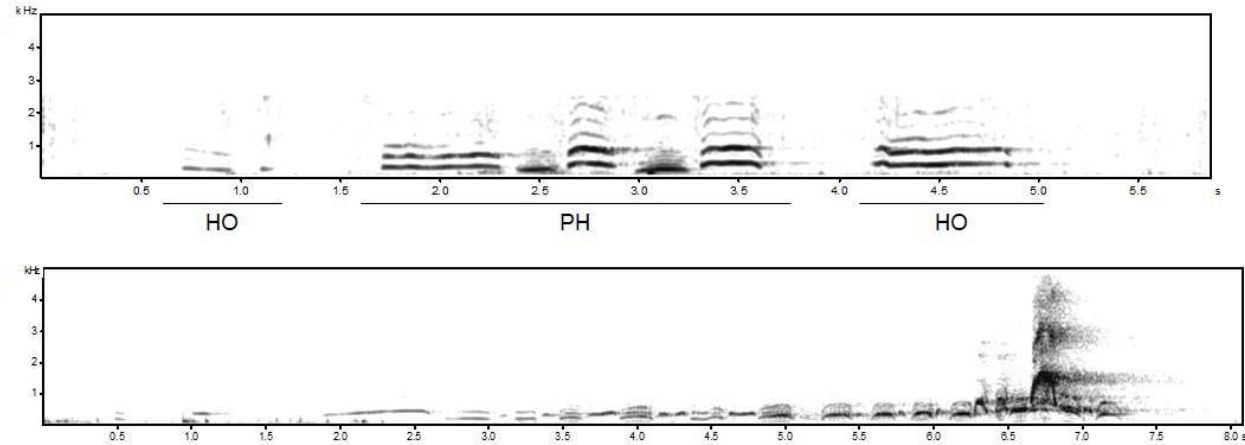
But Desai et al. (2022) fail to replicate in Gombe National Park

- Substantial inter-individual differences, small sample sizes

Crockford, C., Herbinger, I., Vigilant, L. & Boesch, C. (2004). Wild Chimpanzees Produce Group-Specific Calls: a Case for Vocal Learning? *Ethology*, 110, 221–243.

Desai, N. P., Fedurek, P., Slocombe, K. E., & Wilson, M. L. (2022). Chimpanzee pant-hoots encode individual information more reliably than group differences. *American Journal of Primatology*, 84, e23430.

A lot is not known about call combinations in chimpanzees!



Girard-Buttoz, C., Zaccarella, E., Bortolato, T., Friederici, A. D., Wittig, R. M., & Crockford, C. (2022). Chimpanzees produce diverse vocal sequences with ordered and recombinatorial properties. *Communications Biology*, 5, 410.

Communication in the Last Common Ancestor of chimpanzees, bonobos, and humans?

- Not 2nd order intentional?
- No/minimal use of structure subserving meaning?
- Probably not learned?

But remember Fitch's point: their communication system may underrepresent their cognitive capacities!



Summary of today

- Intentional communication
 - Rare in primates, perhaps present in chimpanzees?
- Structured communication
 - Rare and limited in most primates, common in songbirds
 - Generally structure not subserving meaning
- Learned communication
 - Rare in primates (but perhaps present in chimpanzees?), common in songbirds
 - Relationship between vocal learning and structure?

Next up

- Workshop on chimpanzee gesture comprehension by humans
 - Do we have access to an ancestral ape gestural communication system?
 - **Be prepared, be involved**
- Week 4: no class! Catch up or read ahead.
- Week 5: human evolution, cumulative non-linguistic culture in humans and other animals, inferring language from the archaeological record